

Minimizing Pumping Energy Cost in Real-Time Operations of Water Distribution Systems Using Economic Model Predictive Control

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Abstract: Optimizing pump operations is a challenging task for real-time management of water distribution systems (WDS). With suitable pump scheduling, pumping costs can be significantly reduced. In this research, a novel economic model predictive control (EMPC) framework for real-time management of WDS is proposed. Optimal pump operations are selected based on predicted system behavior over a receding time horizon with the aim to minimize the total pumping energy cost. Time-varying electricity tariffs are considered while all the required water demands are satisfied. The novelty of this framework is to choose the number of pumps to operate in each pump station as decision variables in order to optimize the total pumping energy costs. By using integer programming, the proposed EMPC is applied to a benchmark case study, the Richmond Pruned network. The simulation with an EPANET hydraulic simulator is implemented. Moreover, a comparison of the results obtained using the proposed EMPC with those obtained using trigger-level control demonstrates significant economic benefits of the proposed EMPC. DOI: [10.1061/\(ASCE\)WR.1943-5452.0001399](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001399). © 2021 American Society of Civil Engineers.

Introduction

Water distribution systems (WDS) are critical infrastructure for modern cities. With appropriate operational management, water companies are able to provide water with the desired quantity and quality to all the customers at a reasonable operating cost. Pumping costs constitute a significant proportion of the operating cost for WDS, and they are projected to increase in the future due to an increase in population leading to larger water consumption (IEA 2016). An increasingly important objective in real-time optimal pump operations of WDS is therefore to minimize pumping energy costs. During the last 3 decades, a large amount of research has been carried out on optimal operations of WDS. A comprehensive literature review including over 100 scientific publications from the 1970s until 2017 on the optimization of WDS was conducted by

Mala-Jetmarova et al. (2017). That study revealed that pump operation optimization is one of the two major areas of optimization research related to WDS (the other area being water quality optimization).

WDS operation optimization is a very difficult problem due to a number of challenges, which include but are not limited to the following: (1) the equations governing the hydraulic interactions within the network are **nonlinear**, (2) decision variables can be either **continuous or discrete**, and (3) real WDS typically consist of **a large number of branches, links, loops, and nodes**; in addition, there are new challenges for the operation of modern WDS, such as (4) electricity **prices can change with time**, (5) **disturbances** to the system may occur including degradation of the pipe and pump characteristics, and (6) **the need to enforce constraints on water volumes and pumping flow rates** within a WDS. Consequently, to solve a pump operation problem, a mathematical model of the system is needed taking into account the **nonlinear hydraulic equations** and the **binary (on/off)** nature of pumping decisions. The resulting **mixed-integer dynamic and nonlinear optimization** problem is numerically challenging to solve in real-time. As a result, researchers often make use of a number of simplifying assumptions to reduce the computational requirements, which has resulted in the development of a variety of problem formulations and solution methods over time.

Early studies on WDS operation optimization focused on deterministic approaches, such as dynamic programming (DP) (Sterling and Coulbeck 1975), linear programming (LP) (Giacomello et al. 2013; Kurian et al. 2018), nonlinear programming (NLP) (Brion and Mays 1991; Ormsbee and Reddy 1995), and mixed integer nonlinear programming (MINLP) (Biscos et al. 2003; Bagirov et al. 2013). Recently, metaheuristic algorithms, such as **genetic algorithms** (Paschke et al. 2001; Kazantzis et al. 2002; van Zyl et al. 2004; Wang et al. 2009; Wu et al. 2010, 2012a, b; Blinco et al. 2016), **simulated annealing** (da Conceição Cunha and Sousa 1999; Goldman and Mays 1999), **particle swarm optimization** (Wegley et al. 2000), and **ant colony optimization** (Ostfeld and Tubaltzev 2008; López-Ibáñez et al. 2008) have been used for finding

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open-loop schedules for pump operations. However, the majority of these studies **did not consider real-time pump operations**, where a feedback control loop uses measurements of the current system state to make informed decisions in real time about how to choose system inputs in the future.

A well-known advantage of closed-loop controllers is **robustness** to system modeling errors, although they **require real-time measurements** of system states and **increased online computational resources**. Although a wide range of feedback control algorithms are in use, model predictive control (MPC) has gained increasing popularity over the last decades due to its ability to **explicitly handle system constraints** and deliver improved system performance through the use of a model of the system. It is also readily deployable to **multi-input–multioutput systems**.

MPC was initially developed for the chemical processing industry in the late 1970s (García et al. 1989). It relies on an internal model of the process dynamics to predict system states and outputs given a sequence of control actions over a **finite receding horizon**. An optimization problem is solved at each sampling instant to determine an optimal control sequence that **minimizes deviations of the process states from some predetermined operating setpoints**. MPC **utilizing a quadratic cost on state deviations and control actions** has been applied to many water applications, such as river management and operation of open-channel systems (Nasir et al. 2018, 2019a, b).

Economic MPC (EMPC) is a relatively new extension of MPC, where the **cost function is generally formulated using an arbitrary objective function that captures some economic aspect of process systems to be controlled** (Rawlings et al. 2012; Ellis et al. 2016). Given its superior economic performance, EMPC has found recent applications in various areas, including building energy systems (Ma et al. 2012), gas pipeline networks (Gopalakrishnan and Biegler 2013), electric vehicle charge planning (Halvgaard et al. 2012), inventory management (Subramanian et al. 2014), and wastewater treatment processes (Zeng and Liu 2015). EMPC has been found to be a suitable control strategy for operational management of WDS. An earlier deployment of EMPC for the water distribution problem was proposed by Cembrano et al. (2000), where the simulations only considered a two-pump system model, and locally optimal solutions were found. More recently, this approach has been revisited using **flow-based** and **pressured-based models** of WDS (Wang et al. 2017). Wang et al. (2017) and Salomons and Housh (2020) have suggested the use of EMPC for WDS, but those authors have made simplifications including **approximating the on/off behavior of pumps**, as well as using simplified plant and economic/pump models.

This study proposes a novel EMPC framework for real-time management of WDS, where the **full nonlinear pumping models** and **accurate energy pricing** are taken into account to investigate the benefits of EMPC relative to **trigger-level control**. Within this EMPC framework, a modeling methodology for WDS is introduced, which includes the number of pumps in each pump station **as integer decision variables**. The advantage of this framework is to make sure that accurate economic pumping costs are included in the control system. Based on the objectives of the operational management of WDS, economic cost functions and constraints have been formulated as part of the EMPC optimization problem. By using integer programming to solve the corresponding EMPC optimization problem online with updated feedback information, optimal pump operations can be obtained while minimizing the **pumping costs**. A case study for the Richmond Pruned network is presented here based on the original Richmond network introduced in the study by van Zyl et al. (2004). The closed-loop online simulation results with EMPC connected to the Richmond Pruned

network implemented in the EPANET version 2.2 hydraulic simulator (Rossman 2000). The simulation results also reveal a natural pump operation in view of the electricity prices and the water volumes in storage tanks. A comparison of the results with trigger-level control demonstrate the effectiveness of EMPC.

The remainder of this paper is organized as follows. First, the EMPC framework for real-time pump operations is proposed. Second, the Richmond Pruned case study is described in detail. Then, the closed-loop online simulation results are discussed for six different demand loading cases, and a comparison of the results with those obtained by trigger-level control is presented. Finally, conclusions are drawn and future research directions are suggested.

Economic Model Predictive Control for Optimal Pump Operations

In this section, a novel EMPC framework is systematically described for real-time operational management of WDS. A mathematical modeling methodology is first introduced to find the flow-based model of a WDS. Then, the cost functions and constraints are formulated. The EMPC optimization problem is then presented within this given framework.

WDS Model

To design an EMPC controller, a **lower-order prediction model** of a WDS is required to **describe the system dynamics**. In this research, a flow-based model of WDS is considered, which consists of **T water storage tanks, P_s pump stations with a total of $\bar{n}_j$ parallel pumps located in the j th pump station, and D water demands**. The corresponding variable assignments are given in Table 1.

For each water storage tank $i = 1, \dots, T$, the **variable x_i represents the water depth in the tank**. The water depth in a storage tank as a function of a water volume balance can be written

$$x_i(k+1) = x_i(k) + \frac{\Delta t}{S_i} (q_i^{(in)}(k) - q_i^{(out)}(k)),$$

$$i = 1, \dots, T, k = 0, 1, \dots, N \quad (1)$$

where $q_i^{(in)}$ and $q_i^{(out)}$ = inflow and outflow for the i th water tank; $k = k$ th time step from a total of N time steps; Δt = sampling time; and S_i = area (plan view) of the i th tank. The inflow to a tank is usually determined by the outflows from pump stations, whereas the outflow from a tank is usually determined by the water demands d_s , $s = 1, \dots, D$ from each water demand node.

For each pump station $j = 1, \dots, P_s$, the variable n_j represents the number of parallel pumps that are currently operating, and the **variable q_j represents the total outflow from the j th pump station**.

Table 1. Variable assignment for modeling

Item	Variable	Minimum value	Maximum value
Tank water depth	x_i	0	$\bar{x}_i$
Inflow to the tank	$q_i^{(in)}$	0	—
Outflow from the tank	$q_i^{(out)}$	0	—
Number of pumps operating	n_j	0	$\bar{n}_j$
Outflow from pump station	q_j	0	—
Water demand	d_s	0	—
Subscript indexing the tanks	i	1	T
Subscript indexing the pump stations	j	1	P
Subscript indexing the water demands	s	1	D

The pump flow rate is dependent on downstream hydraulic conditions such as water demand loading and tank water depths. However, because the variations in flow rates due to these factors are relatively small, approximations of the flow rates that only depends on the number of pumps operating is used, i.e., as follows:

$$q_j(k) = \phi_j(n_j(k)), \quad j = 1, \dots, P_s, k = 0, 1, \dots, N \quad (2)$$

where $\phi_j(n_j)$ can be obtained from interpolated experimental or simulated (e.g., EPANET) data, where n_j can take integer values from 0 to $\bar{n}_j$.

In addition, for nodes without storage in a water system, the sum of inflows should be equal to the sum of outflows. This can be written

$$\mathbf{0} = g(\mathbf{u}(k), \mathbf{d}(k)), \quad k = 0, 1, \dots, N \quad (3)$$

where $g(\mathbf{u}(k), \mathbf{d}(k))$ is a vector function. Each element of this vector function corresponds to a node.

In general, the flow-based prediction model for a WDS can be summarized as follows:

$$\mathbf{x}(k+1) = f(\mathbf{x}(k), \mathbf{u}(k), \mathbf{d}(k)), \quad k = 0, 1, \dots, N \quad (4a)$$

$$\mathbf{0} = g(\mathbf{u}(k), \mathbf{d}(k)), \quad k = 0, 1, \dots, N \quad (4b)$$

where $\mathbf{x} = [x_1, \dots, x_T]^\top$, $\mathbf{u} = [n_1, \dots, n_{P_s}]^\top$, and $\mathbf{d} = [d_1, \dots, d_D]^\top$ denote the vectors of tank water depths, the number of pumps operating in each pump station, and the water demands across the whole water network. For complex WDS, Eq. (4a) is obtained from Eqs. (1) and (2), and Eq. (4b) is obtained from the mass balance relationships for the nodes without storage. In this research, the water demands are assumed to be known along a prediction horizon as considered in EMPC by using demand forecasting methods, such as the ones introduced by Wang et al. (2016) and Salomons and Housh (2020).

Cost Functions and Constraints Setup

In the following, cost functions and constraints are defined based on the objectives for real-time operational management of WDS.

The power consumption P for an operating pump is modeled

$$P = \frac{\gamma Q H_p}{\eta_s \eta_m} \quad (5)$$

where γ = specific weight of water (N/m^3); Q = pump flow (m^3/s); H_p = pump head (m); η_s = pump shaft efficiency; and η_m = motor efficiency. From Eq. (5), the energy consumed by a pump operating for a time interval of Δt is

$$E = P \Delta t \quad (6)$$

From Eqs. (5) and (6), the pump head H_p is used to compute the power and energy for a single pump; however, the flow-based model [Eq. (4)] assumes pump energy usage is independent of head (which is a simplification because it ignores any relationship to the state $\mathbf{x}$). The energy consumed in the j th pump station is therefore approximated as $E_j = \psi_j(n_j)$. Taking into account the time-varying electricity prices from the tariff database, the pumping energy cost can be computed at each sampling interval for the different prices. For the j th pump station, given a time-varying

electricity price at time step k as $\alpha_j(k)$, the total pumping energy cost can be expressed

$$\begin{aligned} \ell_j(n_j(k), \alpha_j(k)) &= \alpha_j(k) \psi_j(n_j(k)), \\ j &= 1, \dots, P_s, k = 0, 1, \dots, N \end{aligned} \quad (7)$$

where $\psi_j(n_j)$ = function that estimates the energy consumed by n_j parallel pumps at the j th pump station (assuming that all the pumps in a pump station are of the same type).

The total economic cost function for the P_s pump stations as a function of the vector $\mathbf{u}(k)$ of the actual number of pumps operating in each pump station can be stated as follows:

$$\ell_e(\mathbf{u}(k), \alpha(k)) = \sum_{j=1}^{P_s} \ell_j(n_j(k), \alpha_j(k)), \quad k = 0, 1, \dots, N \quad (8)$$

where the subscript e = economic cost function associated with pumps that are operating; and $\alpha(k) = [\alpha_1(k), \dots, \alpha_{P_s}(k)]^\top$.

Although it cannot be directly attributed to the short-term operating costs of the network, excessive pump switching is considered undesirable because it can lead to mechanical degradation and early replacement (with a consequent economic impact on the long-term operation of the network). As a result, an implicit economic cost that is associated with turning pumps on and off is introduced using the following economic penalty:

$$\ell_p(\mathbf{u}(k)) = \|\Delta \mathbf{u}(k)\|_R^2 = \Delta \mathbf{u}(k)^\top R \Delta \mathbf{u}(k), \quad k = 0, 1, \dots, N \quad (9)$$

where $\Delta \mathbf{u}(k) = \mathbf{u}(k) - \mathbf{u}(k-1)$; and $\|\cdot\|_R$ = weighted 2-norm by a positive-definite matrix R . The weighting matrix R may be tuned to account for different types of pumps in different pump stations, or the different network architecture implicitly allowing more switching behavior in some pump stations.

In a WDS, physical limitations on water depths in the tanks and the availability of pumps in a pump station must be taken into account. For the T water tanks in the network, the constraint describing the physical limitations or operational limits for water depths can be formulated by

$$\underline{\mathbf{x}} \leq \mathbf{x}(k) \leq \bar{\mathbf{x}}, \quad k = 0, 1, \dots, N \quad (10)$$

where $\underline{\mathbf{x}}$ and $\bar{\mathbf{x}}$ denote vectors of lower and upper bounds for tank water depths, respectively.

For the P_s pump stations in the network, the constraints on the number of available pumps can be described by

$$\mathbf{0} \leq \mathbf{u}(k) \leq \bar{\mathbf{u}}, \quad k = 0, 1, \dots, N \quad (11)$$

where $\bar{\mathbf{u}} = [\bar{n}_1, \dots, \bar{n}_{P_s}]^\top$ denotes the vector of the available number of parallel pumps in each pump station. Elements of the vector $\bar{\mathbf{u}}$ take on integer values.

Optimization Problem Formulation

According to the preceding discussion, the EMPC controller for the operational management of WDS can be implemented by solving a finite-horizon optimization problem considering a prediction horizon $N > 0$, that is, at each time $k \geq 0$, solve

$$\begin{aligned} &\text{minimize} \\ &\tilde{\mathbf{u}}(k), \dots, \tilde{\mathbf{u}}(k+N-1) \end{aligned} \sum_{t=k}^{k+N-1} (\ell_e(\tilde{\mathbf{u}}(t)) + \ell_p(\tilde{\mathbf{u}}(t))) \quad (12a)$$

subject to S

$$\tilde{\mathbf{x}}(t+1) = f(\tilde{\mathbf{x}}(t), \tilde{\mathbf{u}}(t), \tilde{\mathbf{d}}(t)), \quad t = k, \dots, k+N-1 \quad (12b)$$

$$\mathbf{0} = g(\tilde{\mathbf{u}}(t), \tilde{\mathbf{d}}(t)), \quad t = k, \dots, k+N-1 \quad (12c)$$

$$\underline{\mathbf{x}} \leq \tilde{\mathbf{x}}(t+1) \leq \bar{\mathbf{x}}, \quad t = k, \dots, k+N-1 \quad (12d)$$

$$\mathbf{0} \leq \tilde{\mathbf{u}}(t) \leq \bar{\mathbf{u}}, \quad t = k, \dots, k+N-1 \quad (12e)$$

$$\tilde{\mathbf{x}}(k) = \mathbf{x}(k) \quad (12f)$$

$$[\tilde{\mathbf{d}}(k), \dots, \tilde{\mathbf{d}}(k+N-1)]^\top = [\mathbf{d}(k), \dots, \mathbf{d}(k+N-1)]^\top \quad (12g)$$

where the tilde refers to the predicted variables. Eq. (12a) is the total cost function, Eqs. (12b) and (12c) are the constraints that the future states must obey the plant dynamics of a WDS based on the volume balance model in Eq. (4), Eq. (12d) constrains the water depth in each tank, Eq. (12e) constrains the number of parallel pumps that can be operated in each pump station, Eq. (12f) is the initialization constraint to feed back the current measured water depth $\mathbf{x}(k)$ as the first predicted state $\tilde{\mathbf{x}}(k)$, and Eq. (12g) is the demand forecasts over a prediction horizon N under the assumption of no uncertainty in the demand forecasts.

Algorithm 1.

EMPC of WDS

Set the sampling time Δt , e.g., 1 h;

Obtain the mathematical model of a WDS in Eq. (4) (mass balance relationships);

Obtain the function of $\phi_j(n_j)$ in Eq. (2), e.g., Eq. (15) (the approximation relating the flow to the number of pumps operating in each pump station);

Assign values to parameters: α_j , R , $\underline{\mathbf{x}}$, $\bar{\mathbf{x}}$ and $\bar{\mathbf{u}}$ (electricity prices, weighting matrix in the cost function, lower and upper bounds for the water depths, maximum number of pumps available in each pump station);

Assign the total simulation time k_{end} and the prediction horizon N ;

Initialize the tank depths $\mathbf{x}(0)$;

while $0 \leq k \leq k_{\text{end}}$ (for each control time step, e.g., 1 h in the EMPC) **do**

Obtain a sequence of water demand forecasts $\mathbf{d}(k), \dots, \mathbf{d}(k+N-1)$;

Solve the EMPC optimization problem in Eq. (12) using mixed-integer nonlinear programming;

Select the first value of the EMPC optimal control sequence as the optimal control action as in Eq. (13);

Reset the time step $t_l = 0$ for the EPANET simulation;

Send this optimal control action to the EPANET simulator;

while $t_l \leq \Delta t$ (for each EPANET hydraulic time step, e.g., 5 min) **do**

Run the hydraulic simulation;

end

Record the tank depths $\mathbf{x}(k+1)$ from the EPANET at the end of control time step;

Feed the tank depths $\mathbf{x}(k+1)$ back to the EMPC controller;

$k \leftarrow k+1$;

end

Furthermore, additional constraints may be required in the EMPC optimization in Eq. (11) in order to guarantee closed-loop stability. One option is to use a constraint that forces the system onto a periodic trajectory within the control horizon if water demands are periodic, as discussed by Wang et al. (2018). This assumption of a periodic water demand does not need to hold strictly in practice. Other additional constraints such as terminal constraints or an average performance constraint can also guarantee closed-loop stability, as discussed by Angeli et al. (2012). In place of additional constraints that guarantee stability, longer prediction horizons may be used (Grimm et al. 2005).

From the solutions to the optimization problem in Eq. (12) at any time k , the optimal control action at sampling time k is chosen

$$\mathbf{u}(k) = \tilde{\mathbf{u}}^*(k) \quad (13)$$

where $\tilde{\mathbf{u}}^*(k)$ = first value of the optimal control sequence as determined from solving Eq. (12). Thus, at each time step k , a new optimization in Eq. (12) is carried out for a prediction horizon N , but only the first control input is implemented. At the next time step $k+1$, all the system states are updated using the measured data from the WDS, and the optimization in Eq. (12) is solved again with new updated system states.

The EMPC controller in Eq. (12) provides real-time optimal control action for the management of WDS. The implementation steps are summarized in Algorithm 1. This algorithm uses mixed-integer nonlinear programming. The mass balance relationships are linear whereas the number of pumps is an integer. The relationship between the flow and the number of pumps selected varies nonlinearly. The sampling time for the EPANET simulation may be different from the one for EMPC. For example, the sampling time step for EMPC may be chosen as 1 h whereas the sampling time step for the EPANET simulation may be chosen as 5 min.

Case Study: Richmond Pruned Network

The Richmond water network was taken from a part of the Yorkshire water supply area in the UK described in the study by van Zyl et al. (2004). Its EPANET simulation model can be found at the University of Exeter (2000). To test the proposed EMPC framework, a portion of the original Richmond network, called the Richmond Pruned network, was developed based on the Richmond skeleton model. The layout of this network is shown in Fig. 1. This network consists of one water storage tank, two pump stations (one that contains two parallel pumps, 1A and 2A, and a booster pump

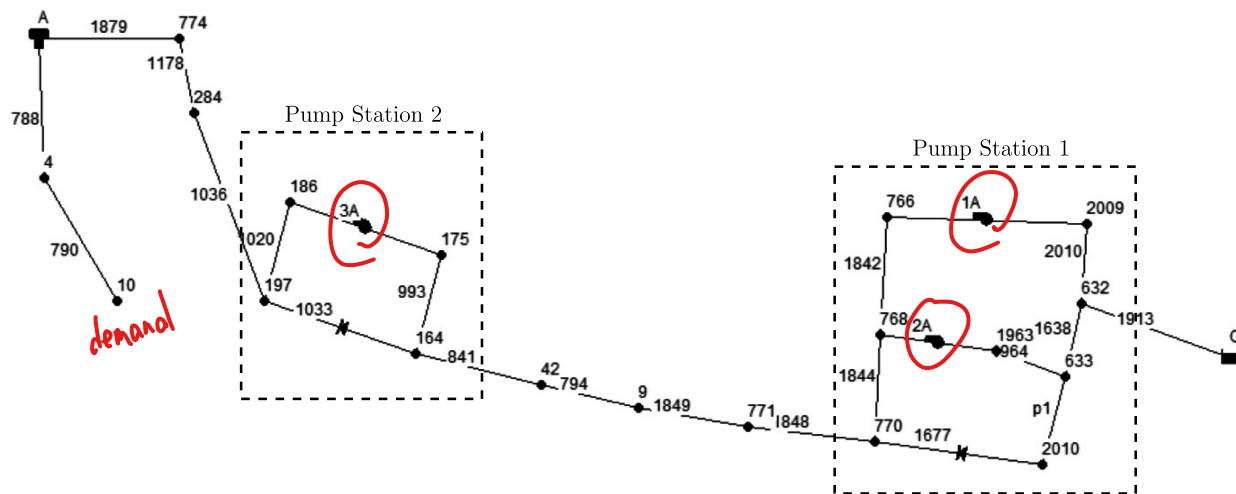


Fig. 1. Layout of the Richmond Pruned network.

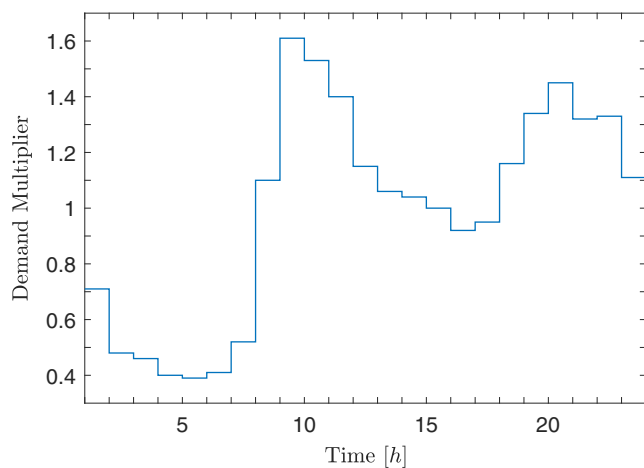


Fig. 2. Demand multipliers for the Richmond Pruned network.

station that contains a single pump, 3A), and one demand sector located at Node 10. The demand multipliers $d_m(k)$ within 24 h are given in Fig. 2, in which the average demand multiplier is 1.0. It is assumed that the demand multipliers are repeated every 24 h. The actual daily demand flow at Node 10, $d_{10}(k)$, is given by the product of a given base demand $\bar{d}_{10}$ and the demand multiplier $d_m(k)$, that is, $d_{10}(k) = \bar{d}_{10}d_m(k)$.

Mathematical Model of the Richmond Pruned Network

The mathematical model of the Richmond Pruned network is introduced in this section. Based on the EPANET results of this

network, the data used to fit power and flow, respectively, as functions of the number of pumps operating for EMPC are reported in Table 2, where the total inflow q_A to Storage tank A and power of the individual pumps for Pump station 1 (PS1) and Pump station 2 (PS2) are shown with different combinations of pump operations defined by the numbers of operating pumps n_1 and n_2 .

Based on the topology of this network as shown in Fig. 1, the function in Eq. (1) can be explicitly written as follows:

$$x(k+1) = x(k) + \frac{\Delta t}{S_A}(q_A(k) - d_{10}(k)) \quad (14)$$

where S_A = plan area of Tank A; and q_A = inflow to Tank A. In Eq. (1), the units for water depth are meters and the units for all the flow variables (inflows, outflows and water demands) are cubic meters per second.

To find the total inflow q_A to Tank A, a polynomial equation based on EPANET derived data can be used in Eq. (2)

$$q_A(k) = c_1n_1(k) + c_2n_1(k)^2 + c_3n_2(k) + c_4n_1(k)n_2(k) \quad (15)$$

where $c_1 = 25.215$; $c_2 = 30.84$; $c_3 = 43.23$; and $c_4 = 57.89$. These parameters are fitted so that Eq. (15) accurately reproduces the flow values obtained from the EPANET as indicated in Table 2.

Cost Functions and Constraints

According to Table 2, the economic cost in Eq. (7) can be approximated by

$$\ell_e(\mathbf{u}(k), \alpha(k)) = \alpha(k)(n_1(k) + n_2(k))p\Delta t, \quad k = 0, 1, \dots, N \quad (16)$$

Table 2. Fitting data for pump inflows to Tank A and pump power values

n_1	n_2	q_A (L/s)	PS1 head (m)	PS2 head (m)	PS1 eff.	PS2 eff.	PS1 power (kW)	PS2 power (kW)	Total power (kW)
0	0	0	—	—	—	—	0	0	0
1	0	25.21	123.88	—	0.66	—	46.32	0	46.32
2	0	30.82	126.92	—	0.44	—	87.03	0	87.03
1	1	43.23	105.48	30.35	0.75	0.60	59.52	21.41	80.93
2	1	57.88	121.63	27.42	0.70	0.70	98.45	22.19	120.64

Note: n_1 = number of pumps operating in PS1; n_2 = number of pumps operating in PS2; and eff. = efficiency.

where $\alpha(k)$ = time-varying price at time k ; Δt = sampling time; and p = constant approximation of total power consumed at both pump stations based on EPANET data. For the Richmond Pruned network, an identical electricity cost price variation was assumed [i.e., the tariff of PS1, $\alpha = 2.41$ pence/kWh (UK) in the off-peak period and $\alpha = 6.79$ pence/kWh in the peak period—the values for PS1 in the original Richmond network are used for both pump stations]. Based on the EPANET results in Table 2, it is obvious that the pump selection of $n_1 = 2$ and $n_2 = 0$ is not a good option compared with the one of $n_1 = 1$ and $n_2 = 0$ because the energy consumed is higher and the pumping flow is lower. Hence, this selection is excluded in the search for optimal pump operations. For the remaining four selections of pumps, the parameter p was set as 40.21 kW based on the values in Table 2.

The weighting matrix R in Eq. (9) is set as follows:

$$R = \begin{bmatrix} 100 & 0 \\ 0 & 50 \end{bmatrix}$$

The order of the terms is chosen to provide a reasonable balance between economic cost of operation, ℓ_e , and the desire to reduce the turning on and off of the pumps to a level commensurate with typical operational practice. The relative weighting between the two terms is an artefact of the asymmetric network used in this case study. The weighting term for PS1 is relatively larger than the one for PS2. This is due to the fact that when Pump 3A in PS2 is turned on, one or two pumps may operate in PS1.

Based on the original Richmond network, the water depth constraint for Tank A (i.e., the capacity of this tank) can be found in the EPANET file as follows:

$$\underline{x} \leq x(k) \leq 3.37 \quad (17)$$

where the minimum water depth was chosen to be $\underline{x} = 1.4$ m for the simulation (allowing for a reserve of approximately 40% of the full storage tank volume).

The control input can be set as $\mathbf{u}(k) = [n_1(k), n_2(k)]^T$ with the constraints

$$[0, 0]^T \leq \mathbf{u}(k) \leq [2, 1]^T \quad (18a)$$

$$n_1(k) \geq n_2(k) \quad (18b)$$

$$n_1(k) - n_2(k) \leq 1 \quad (18c)$$

where Eq. (18a) indicates the available pumps in Pump stations 1 and 2. Eq. (18c) ensures that the pump selection of $n_1 = 2$ and $n_2 = 0$ is not used.

Results

The optimization problem was solved by integer programming implemented with the Yalmip toolbox (Löfberg 2004) and the Artelys Knitro solver (Artelys 2020). The EMPC controller was connected to the EPANET hydraulic simulator. The optimal control action from the EMPC controller was sent to this simulator via the EPANET-MATLAB version R2020a toolkit (Eliades et al. 2016). The EMPC controller was implemented with the sampling time $\Delta t = 1$ h = 3,600 s and the prediction horizon was chosen to be $N = 24$ h. With different base demands at Node 10, the closed-loop simulation results for 4 days (96 h) are preserved and discussed in the following sections.

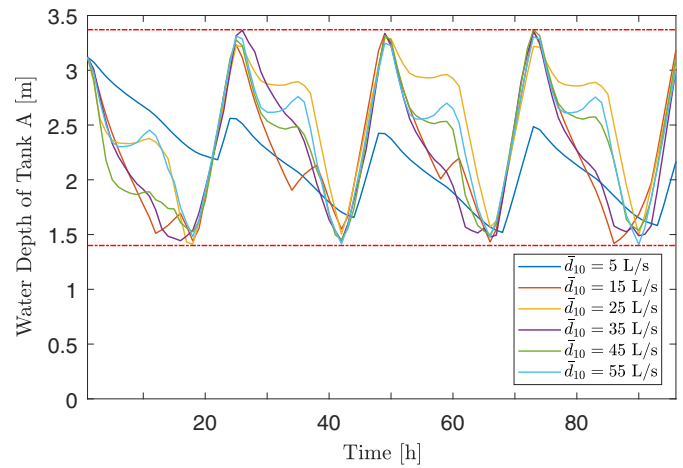


Fig. 3. Water depths in Tank A for different demands at Node 10.

Simulation Results of EMPC with EPANET

To assess the performance of the proposed EMPC, the closed-loop simulations with the Richmond Pruned network in EPANET have been carried out from the same initial water depth $x(0) = 3.12$ m in Tank A. In PS1, Pumps 1A and 2A are of the same type. In simulations, it was arbitrarily chosen to use Pump 2A when $n_1 = 1$. For the demand sector at Node 10, six different water demand loading cases were chosen by setting the base demands as $\bar{d}_{10} = 5, 15, 25, 35, 45, 55$ L/s, respectively. In simulations, it was verified that with the proposed EMPC controller, this network can be operated to satisfy all the demands up to $\bar{d}_{10} = 57.9$ L/s but fails above this level (demands exceed capacity of the pumps and Tank A empties). For the demand loading larger than $\bar{d}_{10} = 57.9$ L/s, no controller or operational strategy can handle such demands because the average demand exceeds maximum pumping capacity. For the cases when $\bar{d}_{10} > 5$ L/s, for the step $t \geq 8$ over the prediction horizon N of Eq. (12), the decision variables $\tilde{\mathbf{u}}(k), \dots, \tilde{\mathbf{u}}(k+7)$ were treated as integer variables, whereas the decision variables $\tilde{\mathbf{u}}(k+8), \dots, \tilde{\mathbf{u}}(k+N-1)$ were treated as continuous variables in order to reduce the computational burden required to find the solution.

For these six cases, the water depth variations of Tank A are shown in Fig. 3. From the same initial water depth, the EMPC optimization results show Tank A fills up and then decreases before arriving at the given minimum depth of $\underline{x} = 1.4$ m. In general, the tank fills up when the electricity tariff is low and empties when it is high. Because the demand multipliers follow a periodic pattern over 24 h, the water depth variations also follow a periodic behavior for each of these six cases after the effect of the initial conditions has subsided.

For $\bar{d}_{10} = 5, 25, 45$ L/s, the resulting pump operations are shown in Figs. 4–6. As shown in Fig. 4, because the base demand is small, only one pump in PS1 is required, and pumping only takes place in the off-peak tariff period as shown in Fig. 4(a). When the demand is increased to $\bar{d}_{10} = 25$ L/s, operation of one pump in PS1 is not enough to provide the required flows and pumping in PS2 is required, but only in the off-peak tariff period, as shown in Fig. 5(b), and some pumping in PS1 is necessary in the peak tariff period. However, only one pump (out of two pumps) in PS1 is operated, and this pump is switched off for some of the time during the peak tariff period. When the demand is increased to $\bar{d}_{10} = 45$ L/s, both pump stations are required to be operated in order to provide enough water. From the optimal solution of

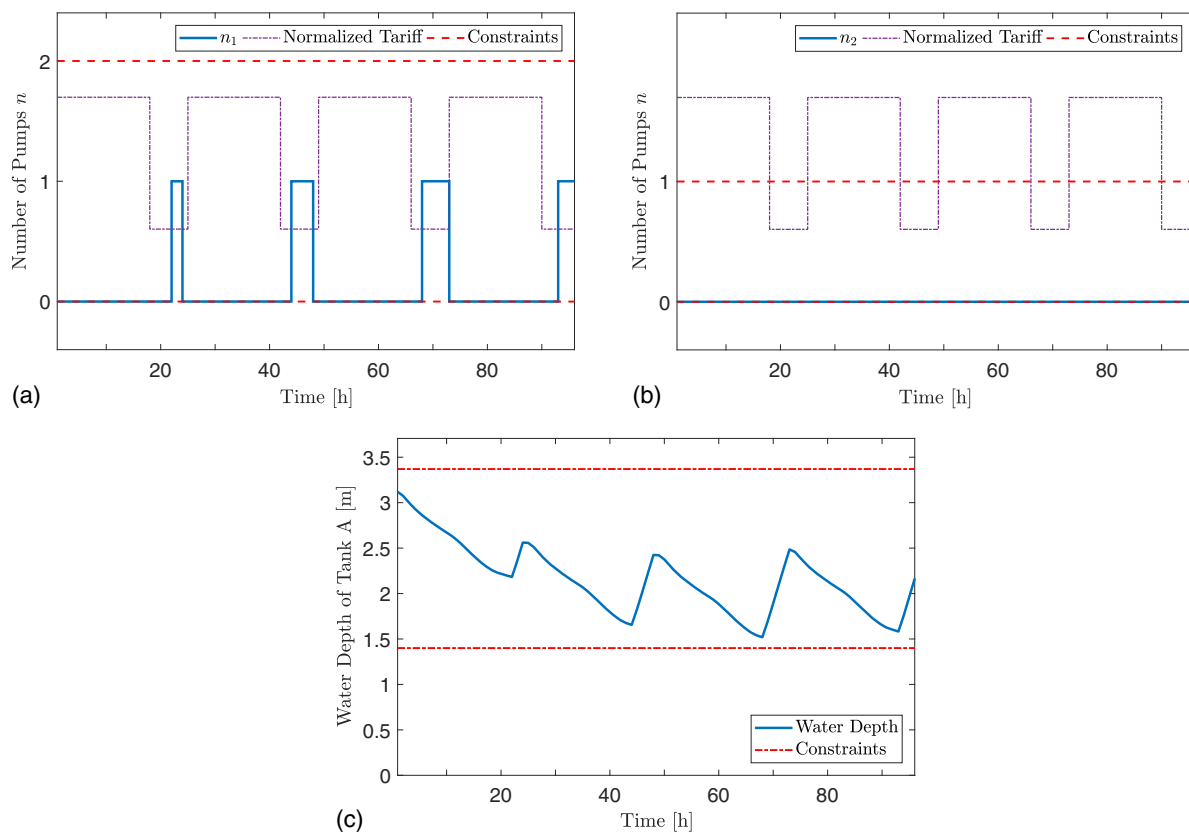


Fig. 4. Optimal pump operations and variation of water depth in Tank A by the EMPC controller with $\bar{d}_{10} = 5$ L/s. Only Pump 2A operates in PS1 in the off-peak tariff period.

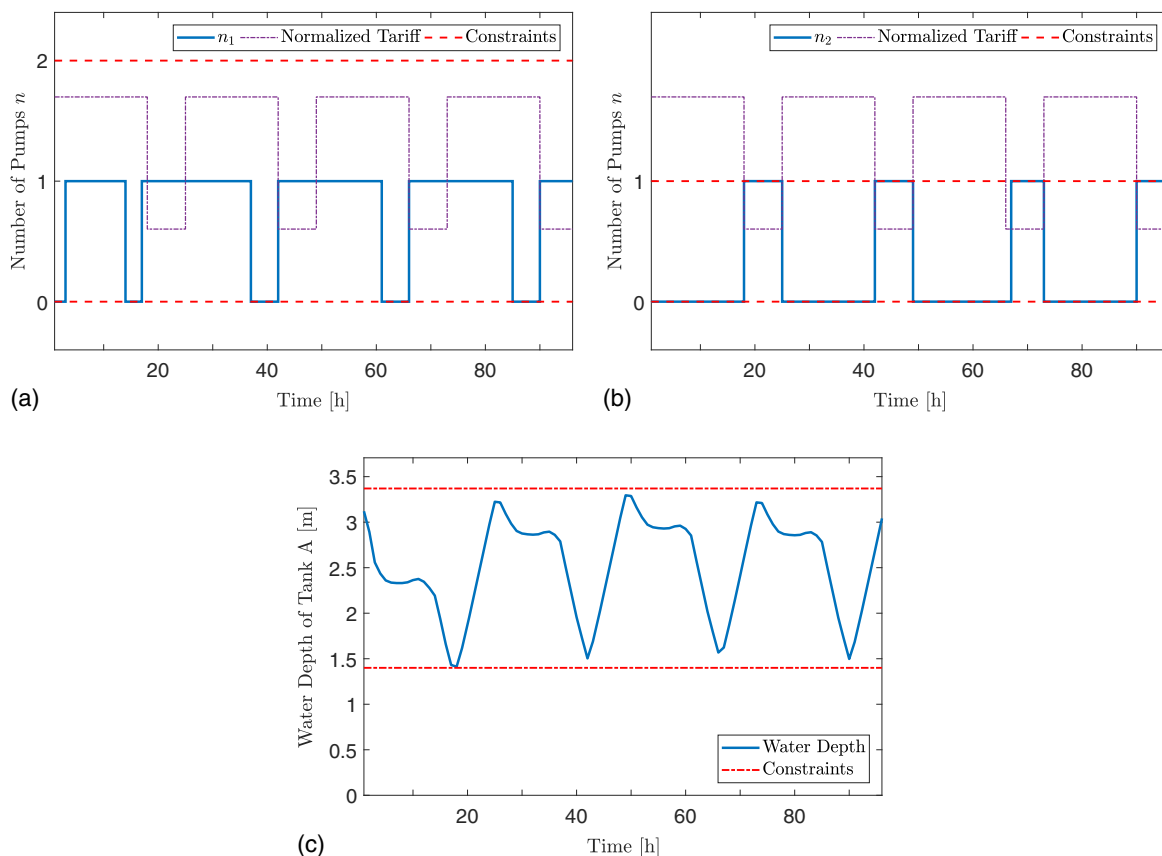


Fig. 5. Optimal pump operations and variation of water depth in Tank A by the EMPC controller with $\bar{d}_{10} = 25$ L/s. Only Pump 2A in PS1 and Pump 3A in PS2 operate. Pump 2A operates all the time in the off-peak tariff period, and Pump 3A only operates in the off-peak tariff period.

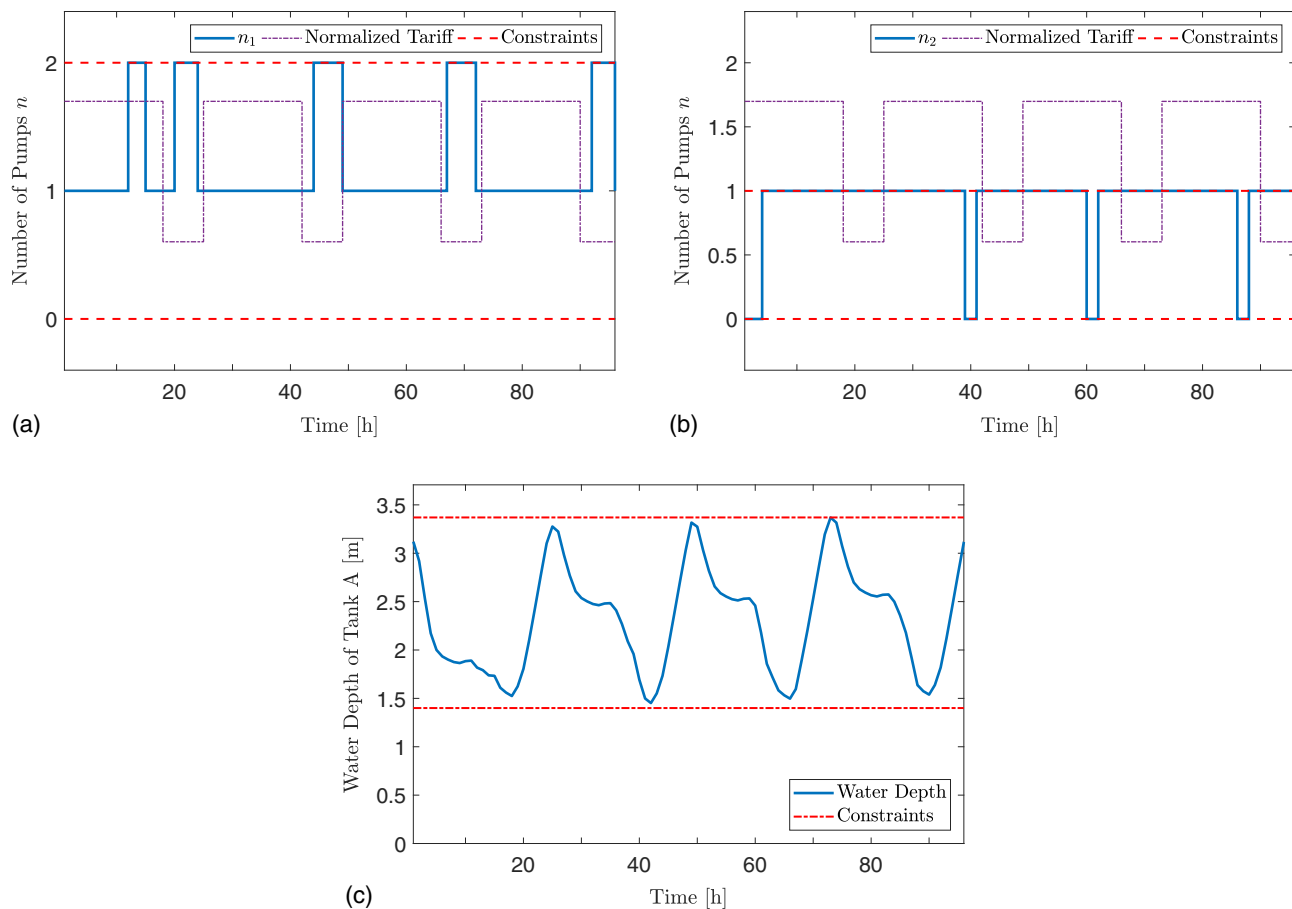


Fig. 6. Optimal pump operations and variation of water depth in Tank A by the EMPC controller with $\bar{d}_{10} = 45$ L/s. All three pumps operate at various times.

EMPC, a single pump in PS1 is operated during the peak tariff period and two pumps in PS1 are operated in the off-peak tariff period, as shown in Fig. 6(a). For PS2, the single pump is needed for most of the time and is only switched off for some of the time during the peak tariff period.

In the cases in Figs. 4 and 5, most of the pumping takes place in the off-peak tariff period, and no unexpected pump switches happened. For the case when $\bar{d}_{10} = 45$ L/s, the optimal pump operations are shown in Figs. 6(a and b) and the water depth variation of Tank A is shown in Fig. 6(c). From the beginning, Tank A drains because the demand is high, and Pump 1A in PS1 is not turned on because it is in the peak tariff period. As the water depth in Tank A gets close to the lower limit, both Pumps 1A and 2A in PS1 are eventually turned on in order to avoid the water depth going below 1.4 m; Pump 1A is only used for a short period of time in the peak

tariff period but used most of time in the off-peak tariff period. Pump 3A in PS2 still operates until around $k = 40$. Because it is still in the peak tariff period and there is enough water in Tank A, Pump 3A is turned off for some time and turned on again in the off-peak period.

To further assess the performance of the proposed EMPC, the total pumped water volume, total consumed energy, and total economic cost are calculated for six water demand loading cases, and the computational results are reported in Table 3. The results of the economic costs are actual pumping energy costs by implementing the chosen pump operations. As the volume of pumped water increases, the consumed energy also increases as well as the total cost. The total cost per unit volume is also computed in Table 3. Because more water is pumped in the peak tariff period, the total cost per unit volume also increases as the water demand increases.

Table 3. Computation results with EMPC for 4-day simulation

$\bar{d}_{10}$ (L/s)	Tot. vol. (m ³)	Total energy (kW·h)	Total cost (£)	Cost per cubic meter (£/m ³)	$\bar{\eta}_{1A}$	$\bar{\eta}_{2A}$	$\bar{\eta}_{3A}$
5	1,398	712	17.16	0.0134	—	0.67	—
15	5,276	2,712	96.69	0.0200	—	0.70	0.60
25	8,693	4,493	207.28	0.0260	0	0.69	0.60
35	12,161	6,427	319.24	0.0286	0.70	0.69	0.64
45	15,580	8,327	444.01	0.0311	0.70	0.73	0.63
55	19,009	10,795	595.67	0.0342	0.70	0.71	0.68

Note: Tot. vol. = total water volume delivered to Tank A.

Table 4. Trigger-level controller setup

Pump	Trigger level ^a : on (m)	Trigger level: off (m)
1A	≤ 2.37	≥ 2.98
2A	≤ 1.40	≥ 3.25
3A	≤ 1.90	≥ 3.11

^aTank A; lower trigger level values for Pumps 2A and 3A were reduced compared with the original Richmond EPANET input file.

Furthermore, the average pump efficiencies for Pumps 1A, 2A, and 3A were obtained from the EPANET simulator and are reported in Table 3. With the proposed EMPC controller, all the average pump efficiencies are around 70%.

Comparison with Trigger-Level Control

To compare the performance of EMPC with a traditional control strategy, the traditional trigger-level control is also applied for

the Richmond Pruned network. The trigger levels for the three pumps are given in Table 4. With $\bar{d}_{10} = 5$ L/s, the results are compared in Fig. 7. At the beginning, the initial tank water depth is $x(0) = 3.12$ m. Based on Table 4, Pump 2A was turned on. Because trigger-level control does not take into account the time-varying electricity price, the pumping flows will occur at any time based on the water depth in Tank A without the regard for the peak and off-peak tariff periods, as shown in Fig. 7(a). In contrast, for EMPC, the pumps are only operated in the off-peak tariff period for $\bar{d}_{10} = 5$ L/s. Consequently, as shown in Figs. 7(b and c), the total pumped water volume and the total energy consumed are similar for both EMPC and trigger-level control, whereas the cost with EMPC is significantly lower than the cost with trigger-level control.

The results obtained from trigger-level control with six different demand loading cases are reported in Table 5. The cost ratio between EMPC and trigger-level control reported in Table 5 is computed by

$$\text{Cost ratio} = \frac{\text{Trigger-level cost per m}^3}{\text{EMPC cost per m}^3} \quad (19)$$

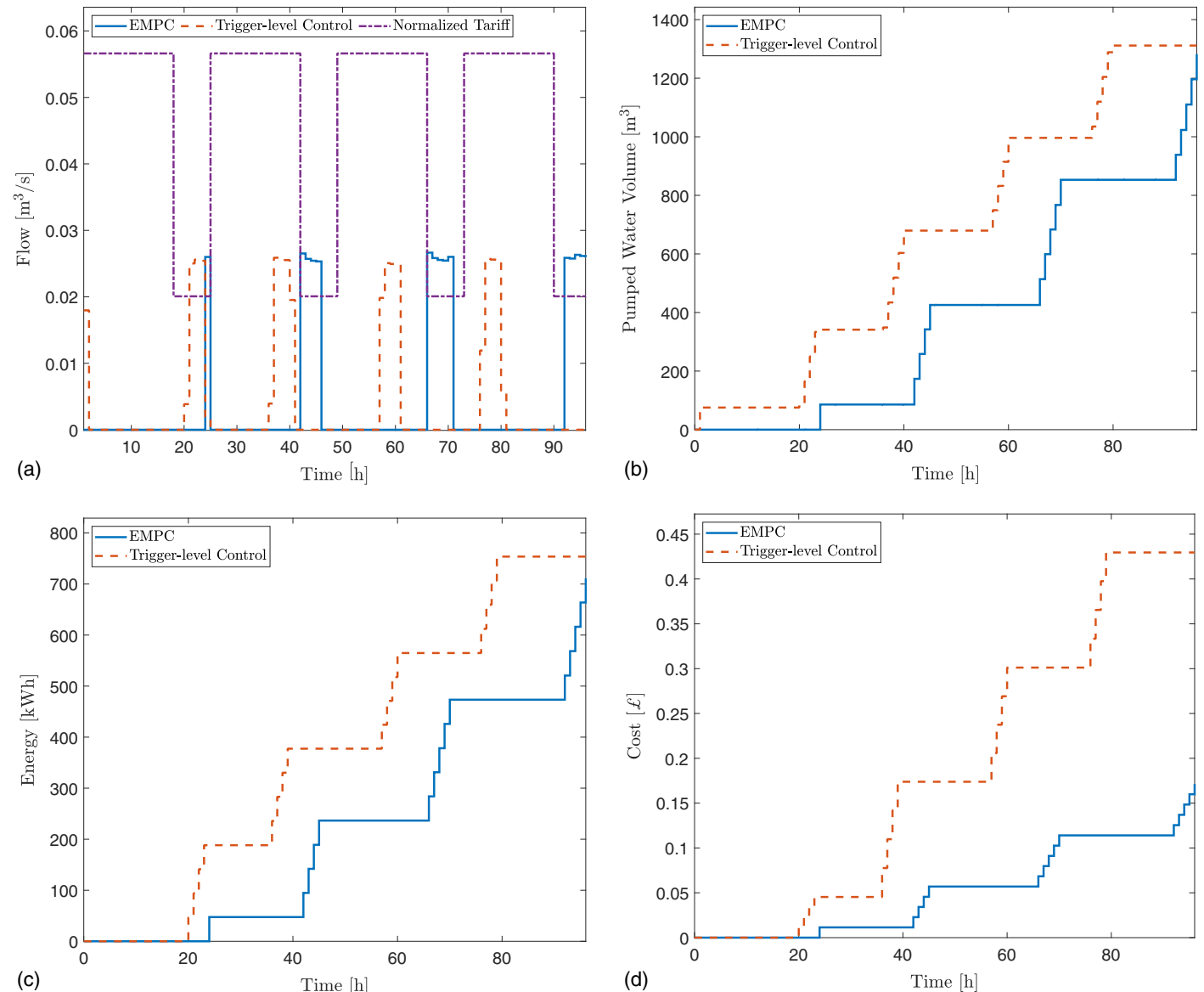


Fig. 7. Comparison of EMPC and trigger-level control with $\bar{d}_{10} = 5$ L/s. For EMPC, Pump 2A operates only in the off-peak tariff period.

Table 5. Computation results with trigger-level control for 4-day simulation

$\bar{d}_{10}$ (L/s)	Tot. vol. (m ³)	Total energy (kWh)	Total cost (£)	Cost per cubic meter (£/m ³)	Cost ratio
5	1,430	753	42.96	0.0328	2.50
15	5,088	2,540	150.17	0.0322	1.55
25	8,534	4,385	242.34	0.0310	1.16
35	11,849	6,477	410.32	0.0378	1.28
45	15,761	8,746	515.52	0.0357	1.16
55	19,093	10,899	618.74	0.0354	1.03

Note: Tot. vol. = total water volume delivered to Tank A; and cost ratio = trigger-level cost per cubic meter/cost per cubic meter.

Note: Cost ratio = trigger-level cost per m³/EMPC cost per m³.

Table 6. Comparison of Day 4 (D4) for EMPC and trigger-level control

$\bar{d}_{10}$ (L/s)	EMPC				Trigger-level control			
	D4 vol. (m ³)	D4 energy (kW · h)	D4 cost (£)	D4 cost per cubic meter (£/m ³)	D4 vol. (m ³)	D4 energy (kW · h)	D4 cost (£)	D4 cost per cubic meter (£/m ³)
5	429	238	5.73	0.0134	346	189	12.84	0.0371
15	1,209	677	22.62	0.0187	1,313	659	38.64	0.0294
25	1,923	1,084	48.78	0.0254	2,216	1,134	62.60	0.0282
35	2,765	1,560	82.20	0.0297	3,024	1,573	102.33	0.0338
45	3,560	2,086	108.54	0.0305	4,043	2,186	128.02	0.0317
55	4,359	2,710	151.47	0.0347	4,285	2,683	154.40	0.0360

Note: D4 vol. = total water volume delivered to Tank A on Day 4.

According to the cost ratio results in Table 5, the optimal pump operations with EMPC has gained an economic benefit between 3% and 250% compared with trigger-level control. The relative benefits are more significant at lower demand flows because there are more opportunities for intelligent scheduling to increase economic efficiency. On the other hand, at the higher demands, because there is less opportunity for improving the economic performance, the utilization of the pumps is very high, and they must be operated during the high-tariff period in order to satisfy demands.

In Table 6, another comparison has been made for results with EMPC and trigger-level control on Day 4. For the six different demand loading cases, the cost and the cost per cubic meter on Day 4 with trigger-level control are more expensive than with EMPC. In particular, for the cases of $\bar{d}_{10} = 5$ and 55 L/s, more water is delivered to Tank A with EMPC, but the cost with EMPC is cheaper than with trigger-level control.

Conclusions

In this study, a novel EMPC framework has been proposed for real-time operational management of WDS. The optimal pump operations are chosen by minimizing pumping energy costs approximated with a flow-based model of WDS. To demonstrate the utility and advantage of this proposed EMPC framework, the Richmond Pruned network case study including three pumps in two pump stations has been used. The closed-loop simulation with an EMPC controller and an EPANET hydraulic simulator has shown that the proposed EMPC framework is effective and efficient in finding a set of optimal pump operations with minimum pumping energy costs taking into account time-varying electricity prices. Less pumping occurs during the peak tariff period and only when it is necessary, whereas more pumps are operated in the off-peak period.

The performance of EMPC has also been compared with the tradition operational control based on trigger-level values. From this comparison, under the smallest demand loading case, the

energy consumption obtained with EMPC and the trigger-level control is similar at the end of the simulation time but the pumping energy is 2.5 times more expensive with trigger-level control than with EMPC. The significantly lower cost obtained with EMPC compared with the trigger-level control is due to the fact that EMPC takes into account the time-varying electricity prices.

Uncertainties exist in both the mathematical model of the WDS and the forecasts of demands and electricity prices. The EMPC strategy has displayed some inherent robustness against uncertainty because it uses a simplified model when carrying out the optimization, whereas the hydraulic simulation of a WDS was carried out using the EPANET simulator. However, robust control strategy for WDS with enhanced robustness against uncertainty is a topic for future research.

As a final point, as the complexity of the network increases, the computational tractability of the proposed algorithm may become challenging. In such circumstances, it may be necessary to consider distributed implementations of the algorithms to show the computational burden among the available computational resources.

Data Availability Statement

Some data, models, or code generated or used during the study are available in the following online repository in accordance with funder data retention policies: <https://github.com/yewangunimelb/watersystem>.

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